



# GlowCore

## SE Project Documentation

Version: 1.00

Date: 20.03.2026

Git Version: 926ad7c

**Project Team:** Cedric Cathomas  
Yoris Kucera  
Dominik Wyss  
Nathanael Fässler

**Project Advisor:** Thomas Bocek

School of Computer Science  
OST Eastern Switzerland University of Applied Sciences

# Contents

|            |                                                      |           |
|------------|------------------------------------------------------|-----------|
| <b>I</b>   | <b>Management Summary .....</b>                      | <b>1</b>  |
| <b>II</b>  | <b>Product Documentation .....</b>                   | <b>3</b>  |
| 1          | Glossary .....                                       | 4         |
| 2          | Requirements .....                                   | 5         |
| 2.1        | Functional Requirements .....                        | 5         |
| 2.1.1      | Actors / Personas .....                              | 5         |
| 2.1.1.1    | Casual Player .....                                  | 5         |
| 2.1.1.2    | Dedicated Player .....                               | 5         |
| 2.1.2      | Use Case Diagram .....                               | 5         |
| 2.1.3      | Use Cases .....                                      | 7         |
| 2.2        | Non-Functional Requirements .....                    | 9         |
| 2.2.1      | NFR Overview .....                                   | 9         |
| 2.2.2      | Priority Matrix .....                                | 10        |
| 2.2.3      | Performance Efficiency .....                         | 11        |
| 2.2.4      | Usability .....                                      | 12        |
| 2.2.5      | Reliability .....                                    | 14        |
| 2.2.6      | Maintainability .....                                | 15        |
| 2.2.7      | Portability .....                                    | 16        |
| 3          | Domain Analysis .....                                | 18        |
| 3.1        | Enemy State Machine .....                            | 18        |
| 4          | Architecture .....                                   | 20        |
| 5          | Quality Measures .....                               | 21        |
| 5.1        | Working Environment .....                            | 21        |
| 5.2        | Testing .....                                        | 21        |
| <b>III</b> | <b>Project Documentation .....</b>                   | <b>22</b> |
| 6          | Initial Project Proposal .....                       | 23        |
| 6.1        | Team Members .....                                   | 23        |
| 6.2        | Availabilities .....                                 | 23        |
| 6.3        | Vision .....                                         | 23        |
| 6.4        | Proposed Realisation .....                           | 24        |
| 7          | Project Plan .....                                   | 25        |
| 7.1        | Resources .....                                      | 25        |
| 7.1.1      | People .....                                         | 25        |
| 7.1.1.1    | Project Management Roles .....                       | 25        |
| 7.1.1.2    | Developer Roles .....                                | 26        |
| 7.1.2      | Time .....                                           | 26        |
| 7.1.3      | Cost .....                                           | 26        |
| 7.2        | Organization .....                                   | 26        |
| 7.2.1      | Meetings .....                                       | 26        |
| 7.2.2      | Planning Tools .....                                 | 27        |
| 7.3        | Schedule .....                                       | 28        |
| 7.3.1      | Work Items .....                                     | 28        |
| 7.3.2      | Long Term Plan .....                                 | 28        |
| 7.3.2.1    | Milestones .....                                     | 28        |
| 7.3.2.2    | Epics .....                                          | 30        |
| 7.3.2.3    | Initial Backlog & MVP .....                          | 31        |
| 7.3.3      | Short Term Plan .....                                | 32        |
| 7.3.3.1    | Sprint 1 (Sprint Zero) 20.02.2026 - 06.03.2026 ..... | 33        |
| 7.4        | Risk Management .....                                | 33        |
| 7.4.1      | Risk Matrix .....                                    | 34        |

|            |                                                    |           |
|------------|----------------------------------------------------|-----------|
| 7.4.2      | Risk Matrix (Mitigated)                            | 35        |
| 8          | Time Tracking Report                               | 36        |
| 8.1        | Time Tracking                                      | 36        |
| 8.2        | Time Statistics                                    | 36        |
| 9          | Personal Reports                                   | 37        |
| 10         | Meeting Minutes                                    | 38        |
| 10.1       | Weekly Scrum / 28.02.2026                          | 38        |
| 10.1.1     | Agenda                                             | 38        |
| 10.1.2     | Todos                                              | 39        |
| 10.2       | Weekly Scrum and Sprint Retrospective / 06.03.2026 | 40        |
| 10.2.1     | Agenda                                             | 40        |
| 10.2.2     | Sprint 1 Retrospective                             | 40        |
| 10.2.2.1   | What Went Well                                     | 40        |
| 10.2.2.2   | What Didn't Go Well                                | 41        |
| 10.2.2.3   | Improvements                                       | 41        |
| 10.2.3     | Todos                                              | 41        |
| 10.3       | Sprint Planning / 06.03.2026                       | 42        |
| 10.3.1     | Sprint goal                                        | 42        |
| 10.4       | Sprint Review (Advisor) / 10.03.2026               | 43        |
| 10.4.1     | Agenda                                             | 43        |
| 10.4.2     | Todos                                              | 43        |
| 10.4.3     | Next dates                                         | 43        |
| 10.5       | Weekly Scrum / 13.03.2026                          | 44        |
| 10.5.1     | Agenda                                             | 44        |
| 10.5.2     | Todos                                              | 45        |
| 10.6       | Weekly Scrum and Sprint Retrospective / DD.MM.YYYY | 46        |
| 10.6.1     | What went well                                     | 46        |
| 10.6.2     | What didn't                                        | 46        |
| 10.6.3     | Todos                                              | 46        |
| 10.6.4     | Retro decisions                                    | 46        |
| 10.7       | Sprint Planning / DD.MM.YYYY                       | 47        |
| 10.7.1     | Sprint goal                                        | 47        |
| 10.7.2     | Selected backlog (commitment)                      | 47        |
| 10.7.3     | Risks / Dependencies                               | 47        |
| 10.7.4     | Todos                                              | 47        |
| <b>IV</b>  | <b>Bibliography</b>                                | <b>49</b> |
| <b>V</b>   | <b>List of Figures</b>                             | <b>50</b> |
| <b>VI</b>  | <b>List of Tables</b>                              | <b>51</b> |
| <b>VII</b> | <b>Code Listings</b>                               | <b>53</b> |

# **Part I**

# **Management Summary**

# Management Summary

## **Part II**

# **Product Documentation**

# 1 Glossary

| Term           | Definition                                                                                                                                                                                  |
|----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Grid           | Refers to the 2D Grid that divides the GlowCore world into rows and columns.                                                                                                                |
| Tile           | A Tile is a single unit of space in the grid and has the form of a square.                                                                                                                  |
| Node           | A Node refers to an object that is occupying a Tile. Not every Tile has a Node, but every Node belongs to a Tile.                                                                           |
| World Item     | World Items are items that can be picked up by the player and physically exist somewhere in the GlowCore world. In contrast to Nodes they do not belong to a Tile and do not have a hitbox. |
| Inventory Item | Inventory Items refer to the items that the player possesses. Certain Nodes such as Chests can also hold Inventory Items                                                                    |

**Table 1:** Glossary

# 2 Requirements

## 2.1 Functional Requirements

### 2.1.1 Actors / Personas

Since a game typically only has one primary actor - **The Player** - it makes sense to define Personas that reflect different types of players. The categorization into Casual- and Dedicated Players covers most player types. In the following, the pronoun **he** is used for simplicity, but the Personas are meant to include all players.

#### 2.1.1.1 Casual Player

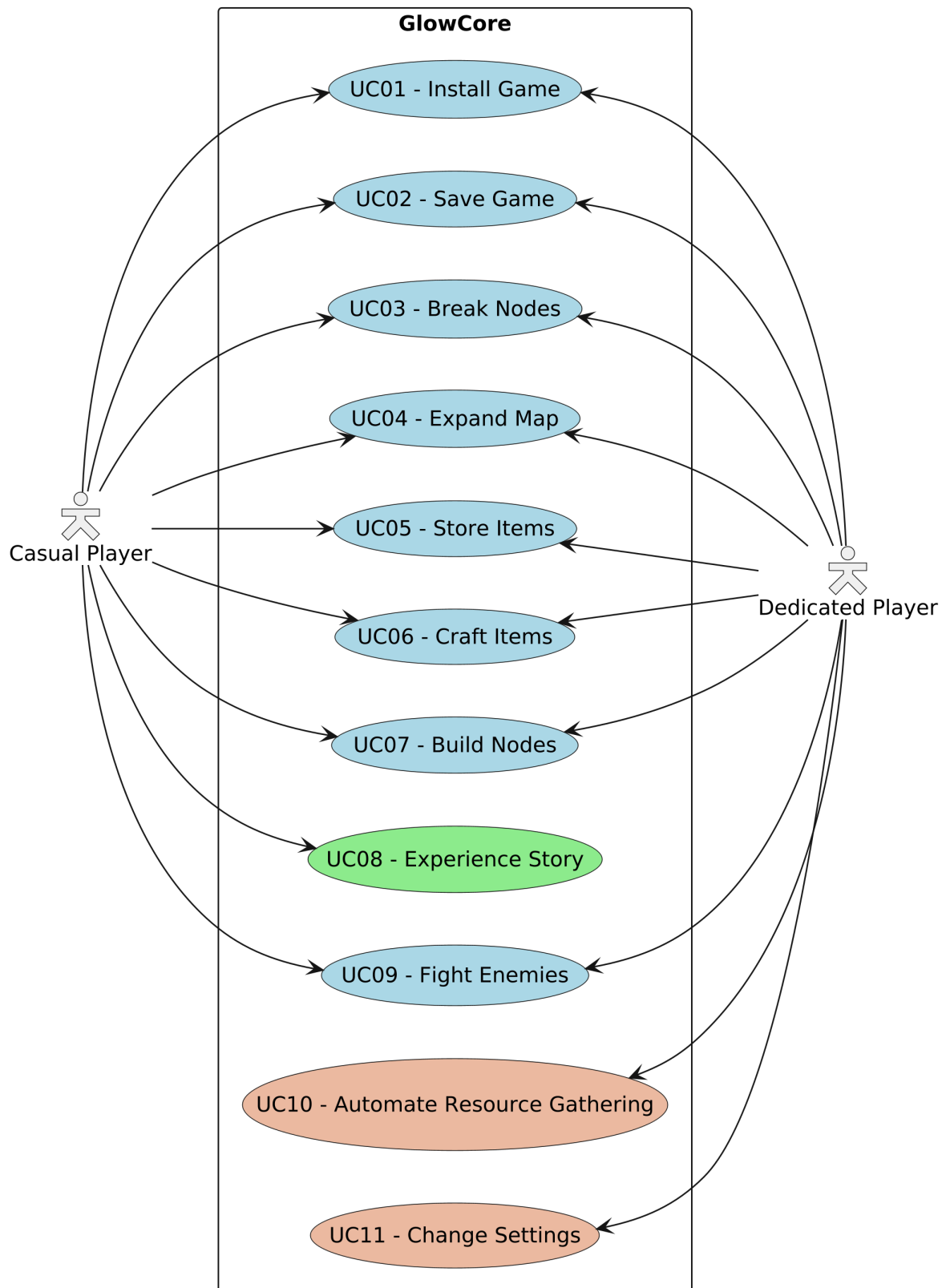
- The Casual Player plays GlowCore infrequently and for short amounts of time. He wants to relax and unwind from his life. He prefers to play the game just for fun without putting effort into it.
- The Casual Player wants to experience the game from beginning to end with a story and explore the world. When he plays the game for the first time, he depends on the game to explain the rules of the game to him without being too overwhelming and tedious. As soon as he finishes the game, he will be done with it, having experienced everything he wanted.
- The Casual Player never changes settings in games, he never feels the need to customize anything. He likes to play the game as it is.

#### 2.1.1.2 Dedicated Player

- The Dedicated Player plays GlowCore in long sessions. He loves games where you build factories and automate everything. He searches for the feeling of progression and finds satisfaction in it. He always searches for a better way to progress and optimize his gameplay.
- The Dedicated Player has already played the game multiple times before, thus he doesn't want to be bored with a long tutorial everytime he starts a new game. He wants to be able to dictate the pacing of the game.
- He doesn't care for the story and lore in games. He doesn't want to read through dialogues and doesn't like long cutscenes.

### 2.1.2 Use Case Diagram

The diagram shows a modified version of the Use Case Diagram. The Personas take the place of the Actors. An arrow indicates that a Persona uses a feature. A missing arrow means that the Persona is not interested in using a feature, although it remains accessible to him.

**Image 1:** Use Case Diagram

## 2.1.3 Use Cases

The Use Cases are described in the brief format, with each Use Case including an appropriately detailed description.

It is essential for the project that all Use Cases that define the **MVP** are implemented. The Use Cases marked with **Priority: Middle** are planned to be implemented, but can be reduced if the MVP takes more time than anticipated. The Use Cases marked with **Priority: Low** will be implemented if enough time is left.

|                    |                                                                                 |
|--------------------|---------------------------------------------------------------------------------|
| <b>ID</b>          | UC01                                                                            |
| <b>Name</b>        | Install Game                                                                    |
| <b>Persona</b>     | Casual Player & Dedicated Player                                                |
| <b>Priority</b>    | High - Required for MVP                                                         |
| <b>Description</b> | The player can install the game. The installation requires no technical skills. |

**Tab. 2:** UC01 - Install Game

|                    |                                                                                                               |
|--------------------|---------------------------------------------------------------------------------------------------------------|
| <b>ID</b>          | UC02                                                                                                          |
| <b>Name</b>        | Save Game                                                                                                     |
| <b>Persona</b>     | Casual Player & Dedicated Player                                                                              |
| <b>Priority</b>    | High - Required for MVP                                                                                       |
| <b>Description</b> | When the player stops playing, the game saves his save state in a file and loads it when he resumes the game. |

**Tab. 3:** UC02 - Save Game

|                    |                                                                                                                                                                                                                                                                                                                                                                       |
|--------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>ID</b>          | UC03                                                                                                                                                                                                                                                                                                                                                                  |
| <b>Name</b>        | Break Nodes                                                                                                                                                                                                                                                                                                                                                           |
| <b>Persona</b>     | Casual Player & Dedicated Player                                                                                                                                                                                                                                                                                                                                      |
| <b>Priority</b>    | High - Required for MVP                                                                                                                                                                                                                                                                                                                                               |
| <b>Description</b> | The player can break Nodes in the world with his hands or with tools. The usage of tools e.g. an axe to break a tree, increases the speed significantly. When the Nodes break, they drop items depending on the type of Node. E.g. a Tree drops a Wood-Item. Certain Nodes that are built by the player themselves such as a workbench drop themselves upon breaking. |

**Tab. 4:** UC03 - Break Nodes

|                    |                                                                                                                                                                                                                    |
|--------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>ID</b>          | UC04                                                                                                                                                                                                               |
| <b>Name</b>        | Expand Map                                                                                                                                                                                                         |
| <b>Persona</b>     | Casual Player & Dedicated Player                                                                                                                                                                                   |
| <b>Priority</b>    | High - Required for MVP                                                                                                                                                                                            |
| <b>Description</b> | The player is confined to only move on tiles that are unlocked at a given point. The player can drop off items at the GlowCore, whereupon the map expands. The player can then move into and explore the new area. |

**Tab. 5:** UC04 - Expand Map

|                    |                                                                                                                                                                                          |
|--------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>ID</b>          | UC05                                                                                                                                                                                     |
| <b>Name</b>        | Store Items                                                                                                                                                                              |
| <b>Persona</b>     | Casual Player & Dedicated Player                                                                                                                                                         |
| <b>Priority</b>    | High - Required for MVP                                                                                                                                                                  |
| <b>Description</b> | Items that the player picks up are stored in his inventory. The player's inventory has a limited size. To organize items, the player can move items into chests and take them out again. |

**Tab. 6:** UC05 - Store Items

|                    |                                                                                                                                                                                                                                                                                                                             |
|--------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>ID</b>          | UC06                                                                                                                                                                                                                                                                                                                        |
| <b>Name</b>        | Craft Items                                                                                                                                                                                                                                                                                                                 |
| <b>Persona</b>     | Casual Player & Dedicated Player                                                                                                                                                                                                                                                                                            |
| <b>Priority</b>    | High - Required for MVP                                                                                                                                                                                                                                                                                                     |
| <b>Description</b> | The player can craft items into other items with predefined recipes. Without a workbench the player is limited to a few simple recipes. The player can interact with a workstation such as a workbench to unlock more crafting recipes. Workstations such as an oven require additional items: e.g. Coal to smelt an Ingot. |

**Tab. 7:** UC06 - Craft Items

|                    |                                                                                                                                                                                          |
|--------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>ID</b>          | UC07                                                                                                                                                                                     |
| <b>Name</b>        | Build Nodes                                                                                                                                                                              |
| <b>Persona</b>     | Casual Player & Dedicated Player                                                                                                                                                         |
| <b>Priority</b>    | High - Required for MVP                                                                                                                                                                  |
| <b>Description</b> | The player can place certain items onto the world, where they become Nodes. Each Tile can hold only one Node, and the player is not allowed to build a Node on an already occupied Tile. |

**Tab. 8:** UC07 - Build Nodes

|                    |                                                                                                                                                                                                                                                                                                                                                                |
|--------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>ID</b>          | UC08                                                                                                                                                                                                                                                                                                                                                           |
| <b>Name</b>        | Experience Story                                                                                                                                                                                                                                                                                                                                               |
| <b>Persona</b>     | Casual Player                                                                                                                                                                                                                                                                                                                                                  |
| <b>Priority</b>    | Low                                                                                                                                                                                                                                                                                                                                                            |
| <b>Description</b> | The player can experience a story while playing the game. When the game starts there is an intro sequence. There is an end to the game, at which point a cutscene is played. The game contains small narrative hints that fit into the story conveyed by the GlowCore-World, including the acoustic, the visual art style and the overall gameplay experience. |

**Tab. 9:** UC08 - Experience Story

|                    |                                                                                                                                                                                                                                                                          |
|--------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>ID</b>          | UC09                                                                                                                                                                                                                                                                     |
| <b>Name</b>        | Fight Enemies                                                                                                                                                                                                                                                            |
| <b>Persona</b>     | Casual Player & Dedicated Player                                                                                                                                                                                                                                         |
| <b>Priority</b>    | Middle                                                                                                                                                                                                                                                                   |
| <b>Description</b> | The player can engage in combat with enemies. Defeated enemies drop items that are useful for the player. When the enemy hits the player, he loses hitpoints. When the player has no hitpoints left, he loses some of his items and respawns near the center of the map. |

**Tab. 10:** UC09 - Fight Enemies

|                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>ID</b>          | UC10                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| <b>Name</b>        | Automate Resource Gathering                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| <b>Persona</b>     | Dedicated Player                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| <b>Priority</b>    | Middle                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| <b>Description</b> | The player can automate tedious tasks, such as breaking Nodes. He can craft machines that are placed as Nodes and they automatically perform a task e.g. chopping down trees. There is a selection of machines that can be combined to automate more and more. Example: The player places multiple Replanter-Machines in a row which automatically replant trees on the neighbouring tile. The player places an Axe-Machine onto the start of the line, at which point the Axe-Machine chops one tree after another and then moves onto the next tile. To automate it even further, the player places Rotaters on both which cause the Axe-Machine to turn around on contact. |

**Tab. 11:** UC10 - Automate Resource Gathering

|                    |                                                                             |
|--------------------|-----------------------------------------------------------------------------|
| <b>ID</b>          | UC11                                                                        |
| <b>Name</b>        | Change Settings                                                             |
| <b>Persona</b>     | Dedicated Player                                                            |
| <b>Priority</b>    | Middle                                                                      |
| <b>Description</b> | The player can change settings of the game, such as the FOV and the Volume. |

**Tab. 12:** UC11 - Change Settings

## 2.2 Non-Functional Requirements

This chapter documents the Non-Functional Requirements (NFRs) for the GlowCore project, categorized according to the ISO/IEC 25010:2011 quality model. The ISO/IEC 25010 standard defines a comprehensive set of software quality characteristics that serve as a framework for specifying and evaluating system quality. The following categories are used to structure our NFRs:

- **Performance Efficiency** – NFR1XX
- **Usability** – NFR2XX
- **Reliability** – NFR3XX
- **Maintainability** – NFR4XX
- **Portability** – NFR5XX

Each NFR is documented with a description, concrete acceptance criteria, a measurement method, a verification approach, and a priority. The priority consists of two parts:

- **MVP Relevance:**
  - **Required** – must be fulfilled for the MVP.
  - **Optional** – planned for a later stage, not required for MVP delivery.
- **Importance:**
  - **High** – critical for the product.
  - **Medium** – important but not blocking.
  - **Low** – nice to have.

Each NFR table includes a colored status indicator in the top-right corner representing its current verification state:

| Color                                                                               | Status                                               |
|-------------------------------------------------------------------------------------|------------------------------------------------------|
|  | Not yet tested (default)                             |
|  | Passed – all acceptance criteria met                 |
|  | Partially met – some criteria passed, others pending |
|  | Failed – acceptance criteria not met                 |

### 2.2.1 NFR Overview

The following table provides a compact overview of all non-functional requirements, their category, priority, verification milestone, and current status. This serves as a quick reference before diving into the detailed NFR descriptions below.

| ID                     | Name                           | Category        | Priority          | Verify at   | Status |
|------------------------|--------------------------------|-----------------|-------------------|-------------|--------|
| <a href="#">NFR101</a> | World Load Time                | Performance     | Required / High   | M08 Alpha   | ■      |
| <a href="#">NFR102</a> | In-Game Transition Time        | Performance     | Required / High   | M08 Alpha   | ■      |
| <a href="#">NFR103</a> | Frame Rate                     | Performance     | Required / High   | M08 Alpha   | ■      |
| <a href="#">NFR104</a> | Save File Size                 | Performance     | Required / Medium | M08 Alpha   | ■      |
| <a href="#">NFR105</a> | Minimal Hardware Req.          | Performance     | Optional / Medium | M09 Beta    | ■      |
| <a href="#">NFR201</a> | Player Restriction Visibility  | Usability       | Required / High   | M08 Alpha   | ■      |
| <a href="#">NFR202</a> | Light Upgrade Progress         | Usability       | Required / High   | M08 Alpha   | ■      |
| <a href="#">NFR203</a> | New Player Learnability        | Usability       | Required / High   | M09 Beta    | ■      |
| <a href="#">NFR204</a> | Input Support                  | Usability       | Optional / Medium | M09 Beta    | ■      |
| <a href="#">NFR205</a> | Game Language                  | Usability       | Optional / Low    | M10 Release | ■      |
| <a href="#">NFR206</a> | Visual Style Consistency       | Usability       | Optional / Medium | M09 Beta    | ■      |
| <a href="#">NFR301</a> | Save File Portability          | Reliability     | Required / High   | M08 Alpha   | ■      |
| <a href="#">NFR302</a> | Game Progress Persistence      | Reliability     | Required / High   | M08 Alpha   | ■      |
| <a href="#">NFR303</a> | Save Data Integrity            | Reliability     | Required / High   | M08 Alpha   | ■      |
| <a href="#">NFR401</a> | Extensibility of Game Systems  | Maintainability | Optional / Medium | M09 Beta    | ■      |
| <a href="#">NFR402</a> | Automated Test Coverage        | Maintainability | Required / Medium | M09 Beta    | ■      |
| <a href="#">NFR501</a> | Multi-Platform Support         | Portability     | Required / Medium | M09 Beta    | ■      |
| <a href="#">NFR502</a> | Engine and Rendering, Pipeline | Portability     | Optional / Low    | M10 Release | ■      |
| <a href="#">NFR503</a> | Display Resolution Support     | Portability     | Optional / Medium | M10 Release | ■      |

**Table 13:** NFR Overview – All Non-Functional Requirements at a Glance

Verification is tied to project milestones (see [Milestones](#)) rather than fixed dates to ensure NFRs are checked at meaningful delivery checkpoints:

- **M08 – Alpha Release (10.04.2026):** All Required / High NFRs must pass. This is the MVP gate.
- **M09 – Beta Release (15.05.2026):** Required / Medium and Optional / Medium NFRs are verified. Performance tuning and playtesting rounds.
- **M10 – Official Release (05.06.2026):** Remaining Optional / Low NFRs are verified. Final polish and compliance check.

## 2.2.2 Priority Matrix

The priority matrix visualizes the distribution of NFRs across the two priority dimensions. This helps identify where the project's quality focus lies and whether the prioritization is balanced.

|               | Required (MVP)                                                             | Optional (Post-MVP)                      |
|---------------|----------------------------------------------------------------------------|------------------------------------------|
| <b>High</b>   | NFR101, NFR102, NFR103<br>NFR201, NFR202, NFR203<br>NFR301, NFR302, NFR303 | –                                        |
| <b>Medium</b> | NFR104, NFR402, NFR501                                                     | NFR105, NFR204, NFR206<br>NFR401, NFR503 |
| <b>Low</b>    | –                                                                          | NFR205, NFR502                           |

**Table 14:** NFR Priority Matrix – MVP Relevance vs. Importance

The matrix shows that the core quality effort is concentrated in the **Required / High** cell (9 NFRs), covering performance, usability, and reliability which are the three pillars that directly impact the player experience in the MVP. The **Optional / Low** cell contains only 2 NFRs that represent project constraints rather than testable quality goals. Priority is on delivering a stable, playable MVP before addressing optional requirements.

### 2.2.3 Performance Efficiency

Performance Efficiency addresses the amount of resources used under stated conditions. For GlowCore, this primarily concerns load times, frame rates, and resource consumption to ensure a smooth and responsive gameplay experience.

|                            |                                                                                                                                                       |  |
|----------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|--|
| <b>ID</b>                  | <b>NFR101</b>                                                                                                                                         |  |
| <b>Description</b>         | World Load Time                                                                                                                                       |  |
| <b>Acceptance Criteria</b> | Existing and newly created worlds must load within 3 seconds when the player presses “Start”.                                                         |  |
| <b>Priority</b>            | Required / High                                                                                                                                       |  |
| <b>Measurement</b>         | Measure elapsed time from pressing “Start” until the world is fully loaded and interactive. Tested on minimum hardware specification.                 |  |
| <b>Verification</b>        | Automated load-time benchmark test across 10 consecutive loads with varying world states (new and existing). All runs must complete within 3 seconds. |  |
| <b>Result</b>              |                                                                                                                                                       |  |

**Table 15:** NFR101 – World Load Time

|                            |                                                                                                                                                 |  |
|----------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|--|
| <b>ID</b>                  | <b>NFR102</b>                                                                                                                                   |  |
| <b>Description</b>         | In-Game Transition Time                                                                                                                         |  |
| <b>Acceptance Criteria</b> | In-game transitions (e.g. walking into a cave) shall not exceed 2 seconds.                                                                      |  |
| <b>Priority</b>            | Required / High                                                                                                                                 |  |
| <b>Measurement</b>         | Measure elapsed time from triggering the transition until the new scene is fully loaded and interactive.                                        |  |
| <b>Verification</b>        | Trigger each scene transition during playtesting and record the load time. Automated scene-transition benchmark covering all transition points. |  |
| <b>Result</b>              |                                                                                                                                                 |  |

**Table 16:** NFR102 – In-Game Transition Time

|                            |                                                                                                                                                    |
|----------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>ID</b>                  | <b>NFR103</b>                                                                                                                                      |
| <b>Description</b>         | Frame Rate                                                                                                                                         |
| <b>Acceptance Criteria</b> | The game must run at a minimum of 60 FPS with 1% lows not dropping below 50 FPS on recommended hardware.                                           |
| <b>Priority</b>            | Required / High                                                                                                                                    |
| <b>Measurement</b>         | Use Unity Profiler or an external frame-time analysis tool to record FPS over a 5-minute gameplay session. Evaluate average FPS and 1% low values. |
| <b>Verification</b>        | Run a standardized gameplay scenario on minimum and recommended hardware. Capture FPS metrics and verify thresholds are met.                       |
| <b>Result</b>              |                                                                                                                                                    |

**Table 17:** NFR103 – Frame Rate

|                            |                                                                                                                                                                                                   |
|----------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>ID</b>                  | <b>NFR104</b>                                                                                                                                                                                     |
| <b>Description</b>         | Save File Size                                                                                                                                                                                    |
| <b>Acceptance Criteria</b> | Save files should remain compact and use a binary serialization format. Under the currently defined maximum expected game progression and world configuration, a save file should not exceed 5MB. |
| <b>Priority</b>            | Required / Medium                                                                                                                                                                                 |
| <b>Measurement</b>         | Measure the file size of the save file after a full gameplay session under the defined maximum expected game progression and world configuration.                                                 |
| <b>Verification</b>        | Create save files at various progression stages and verify the file size remains within the defined limit for the current scope. Confirm that a binary serialization format is used.              |
| <b>Result</b>              |                                                                                                                                                                                                   |

**Table 18:** NFR104 – Save File Size

|                            |                                                                                                                           |
|----------------------------|---------------------------------------------------------------------------------------------------------------------------|
| <b>ID</b>                  | <b>NFR105</b>                                                                                                             |
| <b>Description</b>         | Minimal Hardware Requirements                                                                                             |
| <b>Acceptance Criteria</b> | The game must run on minimal hardware, targeting low-end systems to maximize accessibility.                               |
| <b>Priority</b>            | Optional / Medium                                                                                                         |
| <b>Measurement</b>         | Test the game on a defined minimum hardware baseline and verify all performance NFRs (FPS, load times) are met.           |
| <b>Verification</b>        | Execute a full gameplay session on minimum-spec hardware. Record FPS, load times, and memory usage to confirm compliance. |
| <b>Result</b>              |                                                                                                                           |

**Table 19:** NFR105 – Minimal Hardware Requirements

## 2.2.4 Usability

Usability covers the degree to which the product can be used effectively, efficiently, and satisfactorily. For GlowCore, this includes visual clarity of game mechanics, learnability for new players, and input support.

|                            |                                                                                                                                                                                                        |
|----------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>ID</b>                  | <b>NFR201</b>                                                                                                                                                                                          |
| <b>Description</b>         | Player Restriction Visibility                                                                                                                                                                          |
| <b>Acceptance Criteria</b> | Player restrictions in the game must be clearly communicated through visual indicators such as greyed-out or red-tinted elements. Players must understand restrictions without additional explanation. |
| <b>Priority</b>            | Required / High                                                                                                                                                                                        |
| <b>Measurement</b>         | Conduct playtesting with at least 3 new players. Each player must correctly identify restricted actions without external guidance.                                                                     |
| <b>Verification</b>        | During playtesting sessions, observe whether players attempt restricted actions and whether they understand the visual cues. Document success rate.                                                    |
| <b>Result</b>              |                                                                                                                                                                                                        |

**Table 20:** NFR201 – Player Restriction Visibility

|                            |                                                                                                                                          |
|----------------------------|------------------------------------------------------------------------------------------------------------------------------------------|
| <b>ID</b>                  | <b>NFR202</b>                                                                                                                            |
| <b>Description</b>         | Light Upgrade Progress Indication                                                                                                        |
| <b>Acceptance Criteria</b> | Light upgrade progress must be clearly indicated to the player at all times during gameplay.                                             |
| <b>Priority</b>            | Required / High                                                                                                                          |
| <b>Measurement</b>         | During playtesting, ask players to describe their current upgrade progress. At least 80% must answer correctly.                          |
| <b>Verification</b>        | Conduct playtesting sessions and survey players about their perceived progress. Verify the UI communicates upgrade status unambiguously. |
| <b>Result</b>              |                                                                                                                                          |

**Table 21:** NFR202 – Light Upgrade Progress Indication

|                            |                                                                                                                                                                  |
|----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>ID</b>                  | <b>NFR203</b>                                                                                                                                                    |
| <b>Description</b>         | New Player Learnability                                                                                                                                          |
| <b>Acceptance Criteria</b> | A new player shall understand core mechanics (movement, gathering, upgrading) within 5 minutes without external instructions.                                    |
| <b>Priority</b>            | Required / High                                                                                                                                                  |
| <b>Measurement</b>         | Time how long a new player takes to perform their first gather, move to a new area, and initiate an upgrade without any external help.                           |
| <b>Verification</b>        | Conduct playtesting with at least 3 players who have never seen the game. Measure time to complete core actions. All players must achieve this within 5 minutes. |
| <b>Result</b>              |                                                                                                                                                                  |

**Table 22:** NFR203 – New Player Learnability

|                            |                                                                                                                                                                                |
|----------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>ID</b>                  | <b>NFR204</b>                                                                                                                                                                  |
| <b>Description</b>         | Input Support                                                                                                                                                                  |
| <b>Acceptance Criteria</b> | The game must support both controller and keyboard input methods.                                                                                                              |
| <b>Priority</b>            | Optional / Medium                                                                                                                                                              |
| <b>Measurement</b>         | Test all core gameplay actions with both a standard controller (e.g. Xbox controller) and keyboard. All actions must be executable with both input methods.                    |
| <b>Verification</b>        | Perform a full gameplay session using only controller, then only keyboard. Verify all interactions, menus, and gameplay mechanics are fully accessible with each input method. |
| <b>Result</b>              |                                                                                                                                                                                |

**Table 23:** NFR204 – Input Support

|                            |                                                                                                    |
|----------------------------|----------------------------------------------------------------------------------------------------|
| <b>ID</b>                  | <b>NFR205</b>                                                                                      |
| <b>Description</b>         | Game Language                                                                                      |
| <b>Acceptance Criteria</b> | All in-game text, UI elements, and instructions must be in English.                                |
| <b>Priority</b>            | Optional / Low                                                                                     |
| <b>Measurement</b>         | Review all in-game text and UI elements for non-English content.                                   |
| <b>Verification</b>        | Manual review of all text-containing assets and UI screens to confirm English language throughout. |
| <b>Result</b>              |                                                                                                    |

**Table 24:** NFR205 – Game Language

|                            |                                                                                                                                  |
|----------------------------|----------------------------------------------------------------------------------------------------------------------------------|
| <b>ID</b>                  | <b>NFR206</b>                                                                                                                    |
| <b>Description</b>         | Visual Style Consistency                                                                                                         |
| <b>Acceptance Criteria</b> | All visual assets must follow the low-poly pastel art style consistently throughout the game.                                    |
| <b>Priority</b>            | Optional / Medium                                                                                                                |
| <b>Measurement</b>         | Visual review of all assets against the defined art style guide. No asset shall deviate from the low-poly pastel aesthetic.      |
| <b>Verification</b>        | Conduct an art review session where all in-game assets are compared against reference material. Flag and correct any deviations. |
| <b>Result</b>              |                                                                                                                                  |

**Table 25:** NFR206 – Visual Style Consistency

## 2.2.5 Reliability

Reliability addresses the degree to which a system performs specified functions under specified conditions for a specified period of time. For GlowCore, this is critical for save data integrity and game stability.

|                            |                                                                                                                                                                                           |
|----------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>ID</b>                  | <b>NFR301</b>                                                                                                                                                                             |
| <b>Description</b>         | Save File Portability                                                                                                                                                                     |
| <b>Acceptance Criteria</b> | Each world save must be stored as a single file that can be copied and moved. When placed into the game's save file folder, the save file must be recognized and function without issues. |
| <b>Priority</b>            | Required / High                                                                                                                                                                           |
| <b>Measurement</b>         | Copy a world save file to a different machine or location, place it in the save folder, and verify the game loads it correctly with all progress intact.                                  |
| <b>Verification</b>        | Create a world save file with significant progress. Copy it to a new installation. Load the game and verify all progress, items, and world state for that world are preserved.            |
| <b>Result</b>              |                                                                                                                                                                                           |

**Table 26:** NFR301 – Save File Portability

|                            |                                                                                                                                                                                                              |
|----------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>ID</b>                  | <b>NFR302</b>                                                                                                                                                                                                |
| <b>Description</b>         | Game Progress Persistence                                                                                                                                                                                    |
| <b>Acceptance Criteria</b> | Game progress shall not be lost during normal shutdown. Auto-save shall occur at least every 2 minutes or immediately after a light upgrade.                                                                 |
| <b>Priority</b>            | Required / High                                                                                                                                                                                              |
| <b>Measurement</b>         | Verify auto-save triggers by monitoring save file timestamps during gameplay. Confirm no data loss after normal shutdown.                                                                                    |
| <b>Verification</b>        | Play for several minutes, perform a light upgrade, then shut down the game normally. Restart and verify all progress is retained. Additionally, verify auto-save timestamps occur within 2-minute intervals. |
| <b>Result</b>              |                                                                                                                                                                                                              |

**Table 27:** NFR302 – Game Progress Persistence

|                            |                                                                                                                                                                            |
|----------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>ID</b>                  | <b>NFR303</b>                                                                                                                                                              |
| <b>Description</b>         | Save Data Integrity                                                                                                                                                        |
| <b>Acceptance Criteria</b> | Save data must persist correctly across sessions without corruption.                                                                                                       |
| <b>Priority</b>            | Required / High                                                                                                                                                            |
| <b>Measurement</b>         | Perform repeated save/load cycles (minimum 20) and verify data integrity after each cycle by comparing expected vs. actual game state.                                     |
| <b>Verification</b>        | Automated test that saves game state, reloads it, and compares all serialized fields for correctness. Run across multiple sessions and verify zero corruption occurrences. |
| <b>Result</b>              |                                                                                                                                                                            |

**Table 28:** NFR303 – Save Data Integrity

## 2.2.6 Maintainability

Maintainability represents the degree of effectiveness and efficiency with which a product can be modified. For GlowCore, this covers extensibility of game systems and test coverage.

|                            |                                                                                                                                                        |
|----------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>ID</b>                  | <b>NFR401</b>                                                                                                                                          |
| <b>Description</b>         | Extensibility of Game Systems                                                                                                                          |
| <b>Acceptance Criteria</b> | The code must allow adding new LightCore upgrades and new items with their resource nodes without requiring code restructuring.                        |
| <b>Priority</b>            | Optional / Medium                                                                                                                                      |
| <b>Measurement</b>         | A developer unfamiliar with the codebase must be able to add a new upgrade or item by following existing patterns, without modifying core system code. |
| <b>Verification</b>        | Task a team member with adding a new test upgrade and resource node. Measure the time required and verify no structural code changes were necessary.   |
| <b>Result</b>              |                                                                                                                                                        |

**Table 29:** NFR401 – Extensibility of Game Systems

|                            |                                                                                                                                                                 |
|----------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>ID</b>                  | <b>NFR402</b>                                                                                                                                                   |
| <b>Description</b>         | Automated Test Coverage                                                                                                                                         |
| <b>Acceptance Criteria</b> | All systems must be covered by automated tests. The project targets a minimum of 40% code coverage. (This coverage target is relatively common for video games) |
| <b>Priority</b>            | Required / Medium                                                                                                                                               |
| <b>Measurement</b>         | Measure code coverage using the Unity Code Coverage package. Report overall and per-system coverage percentages.                                                |
| <b>Verification</b>        | Run the full automated test suite and generate a coverage report. Verify that overall coverage meets or exceeds 40%.                                            |
| <b>Result</b>              |                                                                                                                                                                 |

**Table 30:** NFR402 – Automated Test Coverage

## 2.2.7 Portability

Portability addresses the degree of effectiveness and efficiency with which a system can be transferred from one environment to another. For GlowCore, this covers platform support and display resolution compatibility.

|                            |                                                                                                                                                        |
|----------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>ID</b>                  | <b>NFR501</b>                                                                                                                                          |
| <b>Description</b>         | Multi-Platform Support                                                                                                                                 |
| <b>Acceptance Criteria</b> | The game must run on Windows, Linux, and Web platforms.                                                                                                |
| <b>Priority</b>            | Required / Medium                                                                                                                                      |
| <b>Measurement</b>         | Build and deploy the game on all three platforms. Launch and complete a full gameplay session on each.                                                 |
| <b>Verification</b>        | Execute the full test suite and a manual gameplay session on Windows, Linux, and a Web browser. Verify all features work identically across platforms. |
| <b>Result</b>              |                                                                                                                                                        |

**Table 31:** NFR501 – Multi-Platform Support

|                            |                                                                                                                                                                                     |
|----------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>ID</b>                  | <b>NFR502</b>                                                                                                                                                                       |
| <b>Description</b>         | Engine and Rendering Pipeline                                                                                                                                                       |
| <b>Acceptance Criteria</b> | The game must run on the Unity engine using the PhysX physics engine and the Universal Render Pipeline (URP).                                                                       |
| <b>Priority</b>            | Optional / Low                                                                                                                                                                      |
| <b>Measurement</b>         | Verify project settings confirm Unity with PhysX and URP are configured as the active engine and rendering pipeline.                                                                |
| <b>Verification</b>        | Review Unity project settings and confirm PhysX is the active physics engine and URP is the active render pipeline. Verify no fallback to built-in pipeline occurs during gameplay. |
| <b>Result</b>              |                                                                                                                                                                                     |

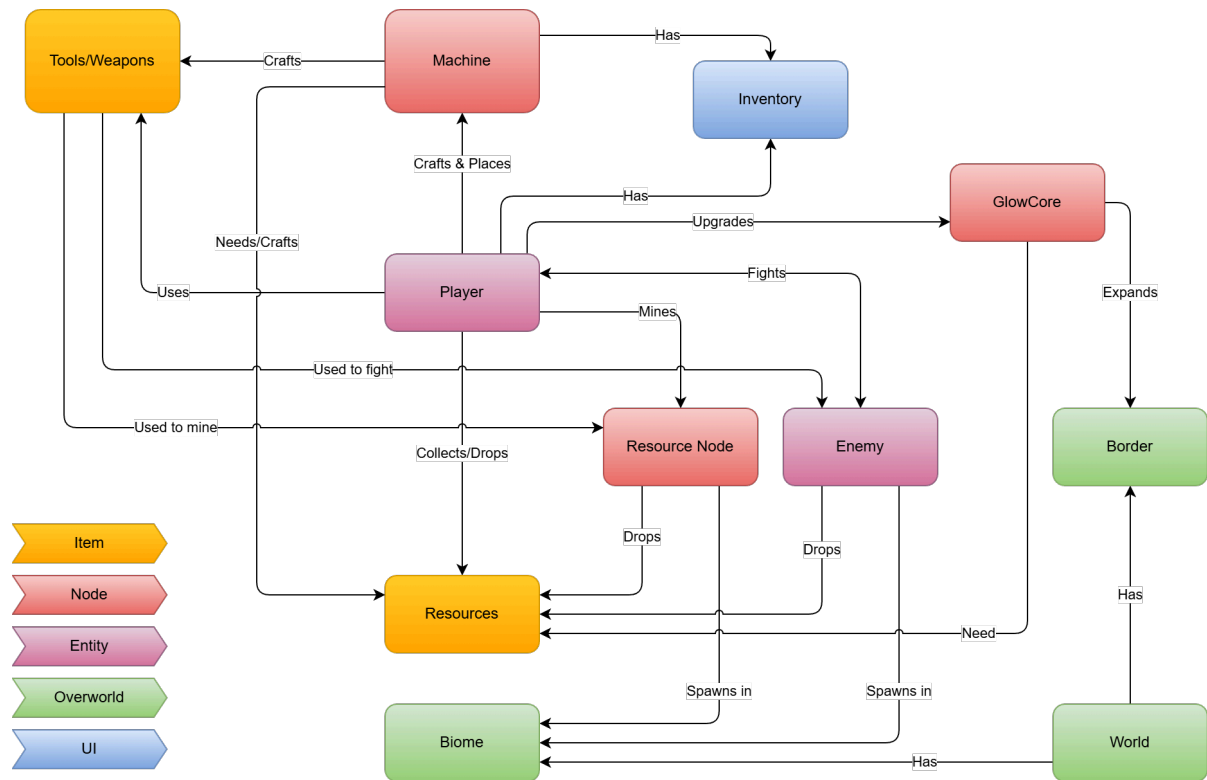
**Table 32:** NFR502 – Engine and Rendering Pipeline

|                            |                                                                                                                                                      |
|----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>ID</b>                  | <b>NFR503</b>                                                                                                                                        |
| <b>Description</b>         | Display Resolution Support                                                                                                                           |
| <b>Acceptance Criteria</b> | The game must support common display resolutions with a 16:9 aspect ratio. Minimum: 1280×720. Recommended: 1920×1080, 2560×1440, and 3840×2160 (4K). |
| <b>Priority</b>            | Optional / Medium                                                                                                                                    |
| <b>Measurement</b>         | Launch the game at each supported resolution and verify the UI scales correctly, no elements are clipped, and the game renders properly.             |
| <b>Verification</b>        | Test the game at all specified resolutions. Take screenshots and verify correct rendering, UI scaling, and absence of visual artifacts.              |
| <b>Result</b>              |                                                                                                                                                      |

**Table 33:** NFR503 – Display Resolution Support

# 3 Domain Analysis

Below is the domain model diagram for the problem domain of our project. Due to the nature of this project, the domain is modeled using an entity-relationship diagram, which captures the key entities and their relationships within the system. The diagram provides a high-level overview of the main components and their interactions, which is essential for understanding the structure of the problem domain.

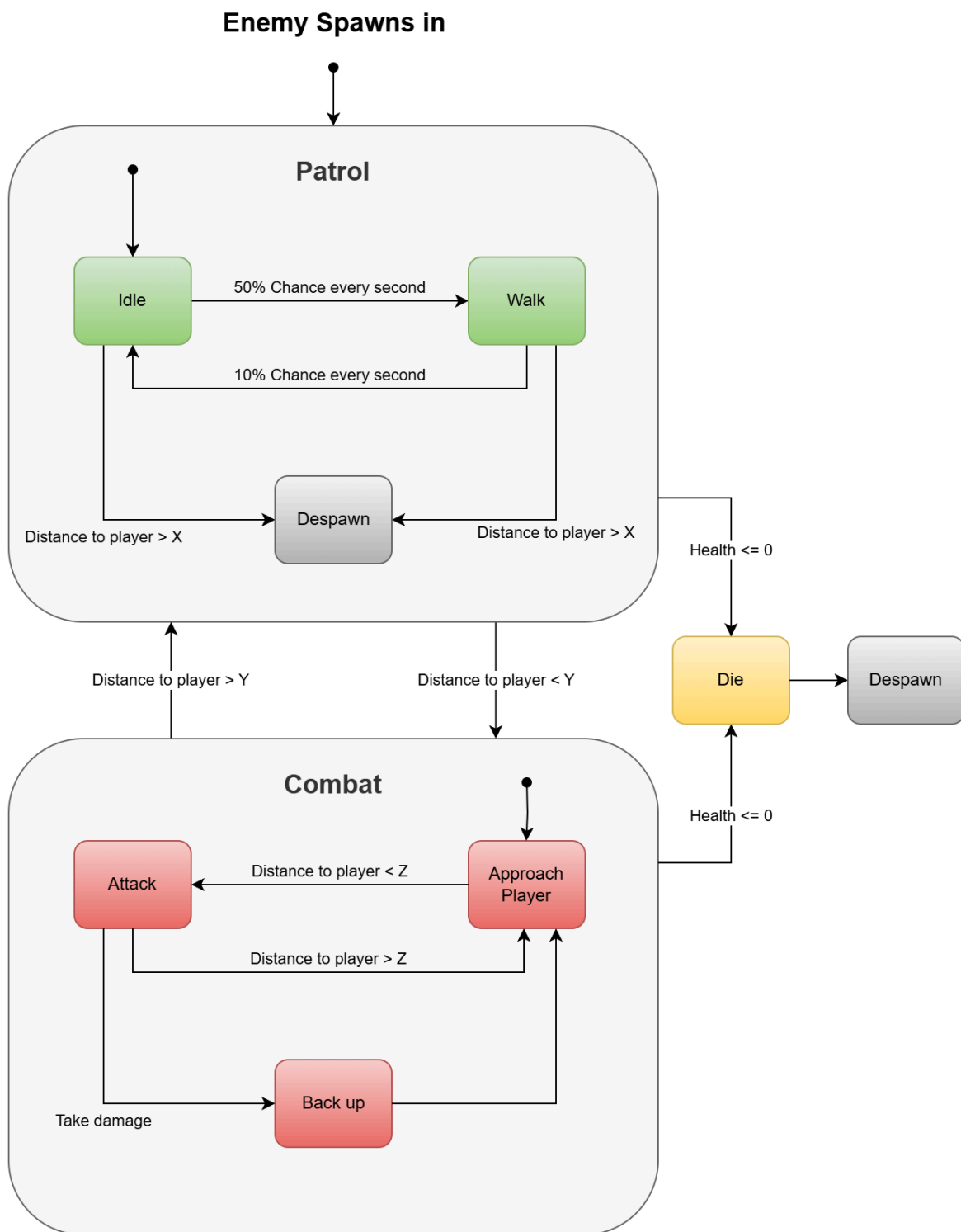


**Image 2: Domain Model Diagram**

One of the key aspects of the domain is the relationship between “Nodes” and “Items”. Some items can be placed in the world which turns them into Nodes. This means that the same concept can have different representations and behaviors depending on the context.

## 3.1 Enemy State Machine

The below diagram represents the state machine for the enemy behavior in our project. It captures the various states an enemy can be in as well as the transitions between those states and the conditions that trigger those transitions.

**Image 3: Enemy State Machine Diagram**

# 4 Architecture

# 5 Quality Measures

## 5.1 Working Environment

## 5.2 Testing

## **Part III**

# **Project Documentation**

# 6 Initial Project Proposal

**Project name:** GlowCore

**Date:** 19.12.2025

## 6.1 Team Members

1. Cedric Cathomas ( [Cedric.Cathomas@ost.ch](mailto:Cedric.Cathomas@ost.ch) )
2. Dominik Wyss ( [Dominik.Wyss1@ost.ch](mailto:Dominik.Wyss1@ost.ch) )
3. Nathanael Fässler ( [Nathanael.Faessler@ost.ch](mailto:Nathanael.Faessler@ost.ch) )
4. Yoris Kucera ( [Yoris.Kucera@ost.ch](mailto:Yoris.Kucera@ost.ch) )

## 6.2 Availabilities

| Time slot   | Mon  | Tue  | Wed | Thu | Fri  |
|-------------|------|------|-----|-----|------|
| 08h00–09h00 | -    | (XR) | -   | -   | (XO) |
| 09h00–10h00 | -    | (XR) | -   | -   | (XO) |
| 10h00–11h00 | (XR) | -    | -   | -   | (XO) |
| 11h00–12h00 | (XR) | -    | -   | -   | (XO) |
| 12h00–13h00 | -    | -    | -   | -   | (XO) |
| 13h00–14h00 | XR   | -    | -   | -   | (XO) |
| 14h00–15h00 | XR   | -    | -   | -   | (XO) |
| 15h00–16h00 | (XR) | XR   | -   | -   | (XO) |
| 16h00–17h00 | (XR) | XR   | -   | -   | (XO) |
| 17h00–18h00 | -    | (XR) | -   | -   | (XO) |
| 18h00–19h00 | -    | (XR) | -   | -   | -    |

**Table 34:** Team availability for review meetings

## 6.3 Vision

GlowCore is a singleplayer progression game focused on resource gathering and gradual world expansion. At the start, the player can only access a small, limited area of the game world, while the rest remains hidden and unreachable. At the center of the world is a special entity that can be upgraded using collected materials. Each upgrade expands the accessible area, revealing new parts of the world and enabling further progression. The core gameplay loop is built around exploring newly unlocked areas, gathering resources, and using them to continuously improve the central entity and push the boundaries of the world outward.

As the world expands, additional gameplay systems deepen the experience. The player can obtain and use different materials, interact with tools, craft items, and face enemies within the expanding world. More advanced mechanics such as machines for processing resources, farming systems, and specialized equipment can be introduced over time to increase complexity and strategic depth.

## 6.4 Proposed Realisation

The project will be implemented using Unity as the primary game engine, using its integrated systems for rendering, physics, input handling, animation, audio, and crossplatform deployment. Development will be done in C# using Unity's component-based architecture (GameObjects, MonoBehaviours, and ScriptableObjects) to structure gameplay logic and systems. Unity's built in physics engine and animation system (Animator) will be used to realise core gameplay and user interaction. Visual assets will be rendered using Unity's Universal Render Pipeline (URP). Individual systems will be tested for robustness using unit tests by method of Unity's own testing framework, based on NUnit. Version control will be handled with Git. The project will be developed within the Unity Editor.

# 7 Project Plan

## 7.1 Resources

### 7.1.1 People

The project team consists of 4 students. Each member contributes approximately 8.6 hours per week and is responsible for tasks of their defined roles.

#### 7.1.1.1 Project Management Roles

| Role            | Description                                                                                                                                                     | Names             |
|-----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|
| Product Owner   | <ul style="list-style-type: none"><li>• Define product vision</li><li>• Manage backlog</li><li>• Prioritize features</li><li>• Validate requirements</li></ul>  | Nathanael Fässler |
| Scrum Master    | <ul style="list-style-type: none"><li>• Facilitate meetings</li><li>• Remove blockers</li><li>• Ensure Scrum process</li><li>• Support team workflow</li></ul>  | Dominik Wyss      |
| Project Manager | <ul style="list-style-type: none"><li>• Schedule planning</li><li>• Resource management</li><li>• Risk control</li><li>• Progress reporting</li></ul>           | Cedric Cathomas   |
| DevOps Engineer | <ul style="list-style-type: none"><li>• CI/CD maintenance</li><li>• Build automation</li><li>• Deployment support</li><li>• Infrastructure management</li></ul> | Yoris Kucera      |

**Table 35:** Project Management Roles

### 7.1.1.2 Developer Roles

| Role                     | Description                                                                                                                                                                        | Names                                                                |
|--------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------|
| Lead Developer           | <ul style="list-style-type: none"> <li>Overall technical coordination</li> <li>Code quality control</li> <li>Architecture decisions</li> <li>Integration management</li> </ul>     | Yoris Kucera                                                         |
| Developer                | <ul style="list-style-type: none"> <li>Gameplay implementation</li> <li>Bug fixing</li> <li>Feature development</li> <li>System integration</li> <li>Writing Test Cases</li> </ul> | Dominik Wyss<br>Yoris Kucera<br>Nathanael Fässler<br>Cedric Cathomas |
| Art Director / 3D Artist | <ul style="list-style-type: none"> <li>Asset creation</li> <li>Scene design</li> <li>Visual consistency</li> <li>Asset optimization</li> </ul>                                     | Nathanael Fässler                                                    |
| UI Architect             | <ul style="list-style-type: none"> <li>UI/UX design</li> <li>Menu systems</li> <li>HUD implementation</li> <li>User interaction flow</li> </ul>                                    | Cedric Cathomas                                                      |
| Graphics Programmer      | <ul style="list-style-type: none"> <li>Shader development</li> <li>Particle effects</li> <li>Performance optimization</li> <li>Visual effects integration</li> </ul>               | Yoris Kucera                                                         |
| Audio Engineer           | <ul style="list-style-type: none"> <li>Sound design</li> <li>Audio integration</li> <li>Music management</li> <li>Audio balancing</li> </ul>                                       | Dominik Wyss                                                         |
| Level Designer           | <ul style="list-style-type: none"> <li>Level layout</li> <li>Gameplay balancing</li> <li>Difficulty tuning</li> <li>Playtesting feedback</li> </ul>                                | Nathanael Fässler<br>Cedric Cathomas                                 |

**Table 36:** Developer Roles

## 7.1.2 Time

The project runs from 16.02.2026 to 12.06.2026 with an estimated effort of 120 hours per person, resulting in a total workload of 480 hours, structured in milestones and sprints.

## 7.1.3 Cost

Most required tools and platform are freely available through free or student licenses. We plan on publishing GlowCore on steam for which steam charges a 100 US-Dollar publishing fee. The cost will be shared by the members.

## 7.2 Organization

### 7.2.1 Meetings

The project meetings follow mainly the Scrum standard. With weekly Scrum meetings replacing daily stand-ups and biweekly sprint planning including retrospectives. Also the Regular advisor meetings to present and discuss our results so far.

| Type                    | Todos                                                                                                                                                                                                       | Frequency                                              |
|-------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------|
| Sprint Review (Advisor) | <ul style="list-style-type: none"> <li>• Present sprint results</li> <li>• Demonstrate features</li> <li>• Discuss open issues</li> <li>• Collect advisor feedback</li> <li>• Document decisions</li> </ul> | Every 2 weeks<br>Tuesday 15:00 - 16:00                 |
| Weekly Scrum            | <ul style="list-style-type: none"> <li>• Report progress</li> <li>• Define next steps</li> <li>• Identify blockers</li> <li>• Update tasks</li> <li>• Support team members</li> </ul>                       | Weekly<br>Friday 14:00 - 15:00                         |
| Sprint Retrospective    | <ul style="list-style-type: none"> <li>• Analyze sprint process</li> <li>• Identify problems</li> <li>• Define improvements</li> <li>• Document lessons learned</li> </ul>                                  | Every 2 weeks<br>Friday 15:00 - 16:00                  |
| Sprint Planning         | <ul style="list-style-type: none"> <li>• Define sprint goals</li> <li>• Select backlog items</li> <li>• Estimate effort</li> <li>• Assign tasks</li> <li>• Update schedule</li> </ul>                       | Every 2 weeks<br>Friday 16:00 - 17:00<br>(after retro) |

**Table 37:** Planned Meetings Overview

## 7.2.2 Planning Tools

| Tool            | Use Case                                                                                                                                                                      |
|-----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| GitHub          | <ul style="list-style-type: none"> <li>• Version control</li> <li>• GitHub Actions (CI/CD)</li> <li>• Pull requests</li> <li>• GitHub notifications via Discord</li> </ul>    |
| Typst           | <ul style="list-style-type: none"> <li>• Project documentation</li> <li>• PDF generation</li> </ul>                                                                           |
| Microsoft Teams | <ul style="list-style-type: none"> <li>• File sharing</li> <li>• Communication</li> </ul>                                                                                     |
| Jira            | <ul style="list-style-type: none"> <li>• Task management</li> <li>• Sprint planning</li> <li>• Backlog tracking</li> <li>• Issue tracking</li> <li>• Time tracking</li> </ul> |
| Miro            | <ul style="list-style-type: none"> <li>• Brainstorming</li> <li>• Diagrams</li> <li>• Project planning</li> </ul>                                                             |
| Discord         | <ul style="list-style-type: none"> <li>• Online meetings</li> <li>• Informal communication</li> <li>• Quick coordination</li> <li>• GitHub activity notifications</li> </ul>  |

**Table 38:** Planning Tools

## 7.3 Schedule

The project follows the OST internal SCRUM+ framework for its overall organization. SCRUM+ combines elements of Scrum with the Rational Unified Process (RUP). Long-term planning is based on the RUP approach, while short-term development is organized using Scrum with two-week iterations.

At the start of each iteration, we estimate all tickets with storypoints. The story **GC-54 Make Player** is the reference story which is worth 5 story points.

### 7.3.1 Work Items

We work with 5 types of Work Items

| Name          | Description                                                            |
|---------------|------------------------------------------------------------------------|
| Epic          | A group of features that we want to implement.                         |
| Story         | Describes a feature that is directly related to the game               |
| Task          | Everything that isn't directly related to the game                     |
| Bug           | Bugs are created when a bug is found and fixed in an unplanned matter. |
| Documentation | Work related to the documentation                                      |

**Table 39:** Work Items

### 7.3.2 Long Term Plan

The Long Term Planning is separated into two categories:

- Milestones: Serve as checkpoint for releases and review meetings.
- Epics: Provide the structure to organize the stories.

#### 7.3.2.1 Milestones

The milestones for the review meetings and for the releases are interleaved and independent.

| Nr  | Milestone Name                        | Date       | Description                                                                                                                                                                                                            |
|-----|---------------------------------------|------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| M01 | Review 1                              | 10.03.2026 | Initial Project Setup                                                                                                                                                                                                  |
| M02 | Review 2                              | 24.03.2026 | Requirements                                                                                                                                                                                                           |
| M03 | Review 3                              | 14.04.2026 | End of Elaboration                                                                                                                                                                                                     |
| M04 | Review 4                              | 28.04.2026 | Quality                                                                                                                                                                                                                |
| M05 | Review 5                              | 12.05.2026 | Architecture                                                                                                                                                                                                           |
| M06 | Project Presentation                  | 12.06.2026 | Final Presentation to SEProject-Advisors and fellow students. Completion of SEProject.                                                                                                                                 |
| M07 | Pre-Alpha Release                     | 20.03.2026 | First playable prototype is ready. Core gameplay loops are implemented. The player can chop trees, pick up wood and expand the GlowCore.                                                                               |
| M08 | Alpha Release                         | 10.04.2026 | MVP is done and working. Core gameplay Mechanics are implemented. Player can craft, build, expand the map and he has an inventory. The player can save his savestate.                                                  |
| M09 | Beta Release                          | 15.05.2026 | More Biomes and Materials are added. There is a combat system with enemies and peaceful mobs.<br>There is a title screen and a settings menu.<br>Automation mechanics are implemented. There are more GlowCore Phases. |
| M10 | Official Release and Final Submission | 05.06.2026 | The game is polished and has no mayor bugs. The game has a beginning and an ending.                                                                                                                                    |

**Table 40:** Milestones

### 7.3.2.2 Epics

| Epic Name                     | Description/Requirements                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|-------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Documentation                 | This epic includes all documentation related work.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| Project Setup                 | This epic contains all work that is done in the Inception-Phase.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| Core Gameplay Loop Mechanics  | <ul style="list-style-type: none"> <li>The player can chop down trees, collect wood and feed it into the GlowCore.</li> <li>The GlowCore levels up and expands the map.</li> <li>The player is restricted to the unlocked map. The grid system is implemented.</li> </ul>                                                                                                                                                                                                                                                                                             |
| Inventory, Crafting, Building | <ul style="list-style-type: none"> <li>Inventory: The player has an inventory and can store a limited amount of items. He can have items in his hands and in his backpack(or similar).</li> <li>Crafting: The player can craft items to other items on different workstations.</li> <li>Building: The player can place objects onto the world-grid.</li> </ul>                                                                                                                                                                                                        |
| Graphics                      | <ul style="list-style-type: none"> <li>The Artstyle of the game ist set.</li> <li>The 3D Models follow a consistent Artstyle.</li> <li>A Shader is implementet.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                            |
| Game Framework & Completeness | <p>This epic contains everything that makes a game complete and feel like a game.</p> <ul style="list-style-type: none"> <li>Intro: When the Game starts the GlowCore Logo is displayed. After the title screen a Cutscene is played that leads into the game.</li> <li>Ending: A ending of the game is defined. When the player finishes the game, a Cutscene is played and the credits roll.</li> <li>UI: A Title Screen is available. In-Game there is a Debug Screen, and a Settings Menu.</li> <li>Gamedata Saving: The player can save his progress.</li> </ul> |
| Big World & GlowCore Phases   | <ul style="list-style-type: none"> <li>World building: The game has a complete world to play in.</li> <li>GlowCore Phases: The GlowCore has multiple phases, all of which need different materials to level up. The GlowCore changes visually with every Level.</li> </ul>                                                                                                                                                                                                                                                                                            |
| Biomes & Materials            | <ul style="list-style-type: none"> <li>The world has multitple biomes: e.g. Forest, Grassland, Mountain, Volcano, Jungle etc.</li> <li>Each biome has unique ressources.</li> <li>There are Caves which you can enter.</li> </ul>                                                                                                                                                                                                                                                                                                                                     |
| Combat                        | <ul style="list-style-type: none"> <li>Enemies can spawn and fight with the player.</li> <li>The player can take damage and can die.</li> <li>A respawn mechanic is set up.</li> <li>There are peacuful mobs.</li> </ul>                                                                                                                                                                                                                                                                                                                                              |
| Resource Automation           | <ul style="list-style-type: none"> <li>The Player can replant trees.</li> <li>The Player can craft machines that automatically gather resources.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                           |
| Playtesting & Release         | <ul style="list-style-type: none"> <li>Playtesting: The game is testet by players that play it for the first time (blind test).</li> <li>Release: The game is released on steam on its own steam page and has a trailer, screenshots and a description.</li> </ul>                                                                                                                                                                                                                                                                                                    |

**Table 41:** Epics

The following image shows the Timeline in Jira as of 04.03.2026.

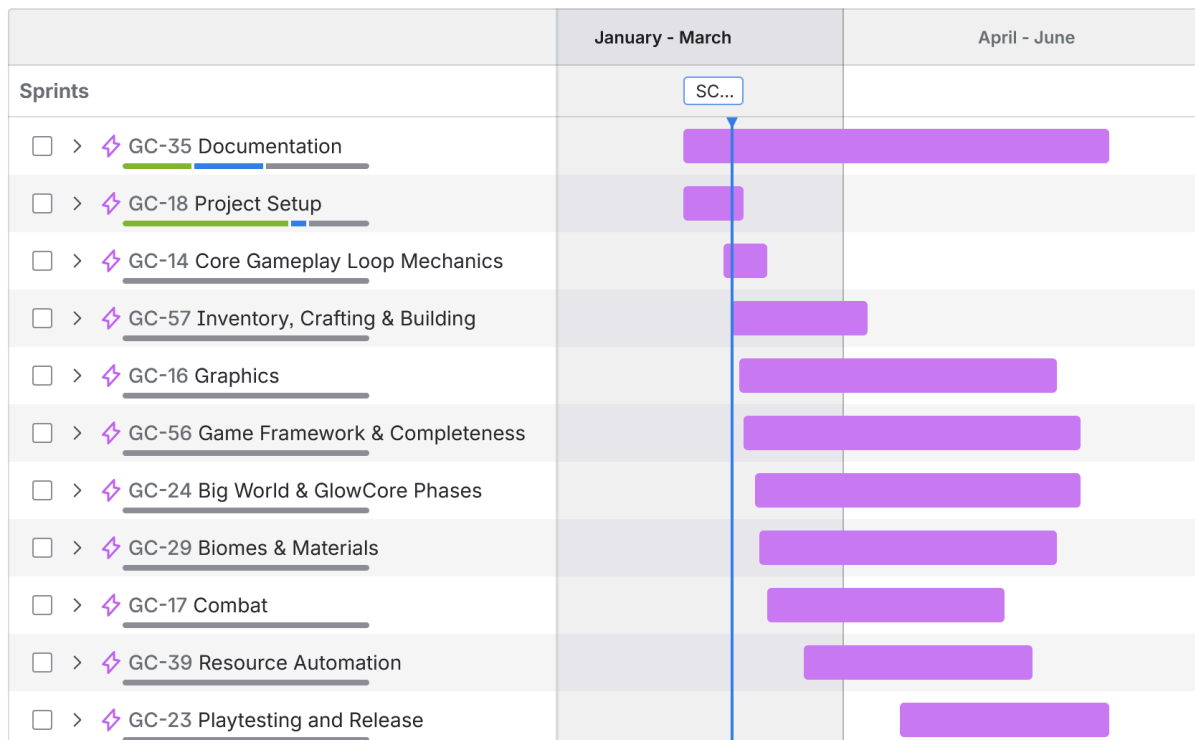


Image 4: Jira Timeline

### 7.3.2.3 Initial Backlog & MVP

The priority of Stories is defined by the ordering in the backlog. As we progress in Agile, we may divide Stories further and add new Stories. The MVP is reached when every Story up to and including **GC-101 Build World** is completed.

The following is the Initial Backlog as of 04.03.2026.

|                                                         |                                        |                      |       |       |                                              |
|---------------------------------------------------------|----------------------------------------|----------------------|-------|-------|----------------------------------------------|
| <input type="checkbox"/> <b>Backlog</b> (44 work items) |                                        |                      |       | 0 0 0 | <input type="button" value="Create sprint"/> |
| GC-54                                                   | There is a player                      | CORE GAMEPLAY LO...  | TO DO | -     |                                              |
| GC-78                                                   | Create Tree-Node 3D Model              | GRAPHICS             | TO DO | -     |                                              |
| GC-95                                                   | Create Wood-Item 3D Model              | GRAPHICS             | TO DO | -     |                                              |
| GC-47                                                   | Trees can be chopped                   | CORE GAMEPLAY LO...  | TO DO | -     |                                              |
| GC-109                                                  | Make GlowCore Phase 1 3D Model         | GRAPHICS             | TO DO | -     |                                              |
| GC-46                                                   | GlowCore expands World                 | CORE GAMEPLAY LO...  | TO DO | -     |                                              |
| GC-69                                                   | Make Inventory-System                  | INVENTORY, CRAFTI... | TO DO | -     |                                              |
| GC-74                                                   | Save Game-Data                         | GAME FRAMEWORK &...  | TO DO | -     |                                              |
| GC-108                                                  | Create Axe 3D Model                    | GRAPHICS             | TO DO | -     |                                              |
| GC-97                                                   | Create Coal 3D Model                   | GRAPHICS             | TO DO | -     |                                              |
| GC-106                                                  | Add Coal Node and Item                 | BIOMES & MATERIALS   | TO DO | -     |                                              |
| GC-96                                                   | Create Stone 3D Model                  | GRAPHICS             | TO DO | -     |                                              |
| GC-93                                                   | Add Stone Node and Item                | BIOMES & MATERIALS   | TO DO | -     |                                              |
| GC-107                                                  | Make Pickaxe 3D Model                  | GRAPHICS             | TO DO | -     |                                              |
| GC-102                                                  | Add Pickaxe                            | BIOMES & MATERIALS   | TO DO | -     |                                              |
| GC-79                                                   | Add Grassland Biome                    | BIOMES & MATERIALS   | TO DO | -     |                                              |
| GC-111                                                  | Make GlowCore Phase 2 3D Model         | GRAPHICS             | TO DO | -     |                                              |
| GC-100                                                  | Add GlowCore Phase 2                   | BIG WORLD & GLOWC... | TO DO | -     |                                              |
| GC-71                                                   | Make Crafting-System                   | INVENTORY, CRAFTI... | TO DO | -     |                                              |
| GC-70                                                   | Make Building-System                   | INVENTORY, CRAFTI... | TO DO | -     |                                              |
| GC-77                                                   | Implement Pixelize-Shader              | GRAPHICS             | TO DO | -     |                                              |
| GC-103                                                  | Add Chest                              | INVENTORY, CRAFTI... | TO DO | -     |                                              |
| GC-101                                                  | Build World                            | BIG WORLD & GLOWC... | TO DO | -     |                                              |
| GC-99                                                   | Add Rock Biome                         | BIOMES & MATERIALS   | TO DO | -     |                                              |
| GC-72                                                   | Game Intro                             | GAME FRAMEWORK &...  | TO DO | -     |                                              |
| GC-73                                                   | Game Ending                            | GAME FRAMEWORK &...  | TO DO | -     |                                              |
| GC-105                                                  | Add Combat Logic                       | COMBAT               | TO DO | -     |                                              |
| GC-124                                                  | Create Enemy 3D Model                  | GRAPHICS             | TO DO | -     |                                              |
| GC-81                                                   | Add Enemy                              | COMBAT               | TO DO | -     |                                              |
| GC-75                                                   | Ingame Debug Settings                  | GAME FRAMEWORK &...  | TO DO | -     |                                              |
| GC-76                                                   | Player Settings-Menu                   | GAME FRAMEWORK &...  | TO DO | -     |                                              |
| GC-125                                                  | Create Peaceful-Mob 3D Model           | GRAPHICS             | TO DO | -     |                                              |
| GC-82                                                   | Add peaceful Mob                       | COMBAT               | TO DO | -     |                                              |
| GC-123                                                  | Make Sword 3D Model                    | GRAPHICS             | TO DO | -     |                                              |
| GC-83                                                   | Add Sword                              | COMBAT               | TO DO | -     |                                              |
| GC-120                                                  | Make Axe-Machine 3D Model              | GRAPHICS             | TO DO | -     |                                              |
| GC-114                                                  | Add Axe-Machine                        | RESOURCE AUTOMAT...  | TO DO | -     |                                              |
| GC-121                                                  | Make Replant-Machine 3D Model          | GRAPHICS             | TO DO | -     |                                              |
| GC-115                                                  | Add Replant-Machine                    | RESOURCE AUTOMAT...  | TO DO | -     |                                              |
| GC-122                                                  | Make Rotater-Block 3D Model            | GRAPHICS             | TO DO | -     |                                              |
| GC-116                                                  | Add Rotater-Block                      | RESOURCE AUTOMAT...  | TO DO | -     |                                              |
| GC-117                                                  | Conduct Playtesting with User          | PLAYTESTING AND R... | TO DO | -     |                                              |
| GC-118                                                  | Release Game on Steam                  | PLAYTESTING AND R... | TO DO | -     |                                              |
| GC-119                                                  | Make Marketing Material for Steam Page | PLAYTESTING AND R... | TO DO | -     |                                              |

Image 5: Inital Backlog

### 7.3.3 Short Term Plan

In a Sprint the priority of the tasks and stories is defined by the ordering.

### 7.3.3.1 Sprint 1 (Sprint Zero) 20.02.2026 - 06.03.2026

The following shows the Jira sprint backlog of the first sprint. Since this was Sprint Zero, not all guidelines were followed initially because they had not yet been established.

| Work                                                                                                                                              | Assignee                                                                                              | Status                                                                                                  |
|---------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------|
|  <del>GG-88</del> Document Long and Short Term Plan              |  Nathanael Fässler   | <span>DONE</span>    |
|  <del>GG-135</del> Document Sprint Planning Meeting Sprint 2     |  Nathanael Fässler   | <span>DONE</span>    |
|  <del>GG-92</del> Setup Contract with OST about Ownership        |  Nathanael Fässler   | <span>DONE</span>    |
|  <del>GG-89</del> Document Meeting Weekly Scrum Sprint 1         |  Nathanael Fässler   | <span>DONE</span>    |
|  <del>GG-4</del> Concept for Code Quality, Linter etc.           |  Yoris Kucera        | <span>DONE</span>    |
|  <del>GG-3</del> Unity Setup                                     |  Yoris Kucera        | <span>DONE</span>    |
|  <del>GG-126</del> Documentation list of tables                  |  Cedric Cathomas     | <span>DONE</span>    |
|  <del>GG-1</del> Risk Specification                              |  Yoris Kucera        | <span>DONE</span>    |
|  <del>GG-91</del> Document Guidelines for code and documentation |  Yoris Kucera        | <span>DONE</span>    |
|  <del>GG-6</del> GitLab Timetracking Research                    |  Dominik Wyss        | <span>DONE</span>    |
|  <del>GG-5</del> GitLab Pipeline Setup                         |  Dominik Wyss      | <span>DONE</span>  |
|  <del>GG-2</del> GitHub Setup                                  |  Yoris Kucera      | <span>DONE</span>  |
|  <del>GG-87</del> Document Meetings Structure                  |  Cedric Cathomas   | <span>DONE</span>  |
|  <del>GG-84</del> Typst documentation changes                  |  Cedric Cathomas   | <span>DONE</span>  |
|  <del>GG-86</del> Document Collaboration as a team             |  Cedric Cathomas   | <span>DONE</span>  |
|  <del>GG-7</del> Typst Documentation Setup                     |  Cedric Cathomas   | <span>DONE</span>  |
|  <del>GG-8</del> People, Cost, Time Documentation              |  Cedric Cathomas   | <span>DONE</span>  |
|  <del>GG-85</del> Roles Documentation                          |  Cedric Cathomas   | <span>DONE</span>  |
|  <del>GG-12</del> Longtermplan & Milestones                    |  Nathanael Fässler | <span>DONE</span>  |
|  <del>GG-9</del> Jira Setup                                    |  Nathanael Fässler | <span>DONE</span>  |
|  <del>GG-112</del> Unity Folder Structure                      |  Nathanael Fässler | <span>DONE</span>  |
|  <del>GG-11</del> Steamworks Registration                      |  Nathanael Fässler | <span>DONE</span>  |
|  <del>GG-13</del> Shorttermplan for first sprint               |  Nathanael Fässler | <span>DONE</span>  |

Image 6: Sprint 1 Backlog

## 7.4 Risk Management

This chapter describes potential risks to the project and how we plan to mitigate them. We identified 8 risks, which are listed in the table below. Additionally, these risks are visualized in two risk matrices, one showing the initial risk assessment and the other showing the risk assessment after mitigation strategies are applied.

| No. | Risk                           | Description                                                                                                                                                                                                                                  | Mitigation Strategy                                                                                                                                                                                                  |
|-----|--------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| R01 | Lack of familiarity with Unity | None of our team members have any significant experience with the Unity Engine. As such, this could heavily impact our development times.                                                                                                    | We made ourselves familiar with the engine before the actual start of the project, to get an idea of the workflows.                                                                                                  |
| R02 | Unity CI/CD Complexity         | We are depending on Unity's build system being robust enough to work in the context of a CI/CD pipeline. We also potentially rely on pre-existing GitHub Actions for Unity, which may not be well-maintained or compatible with our project. | We will set up the CI/CD pipeline very early into the project and did earlier tests. We tried out pre-existing GitHub Actions to make sure they satisfy our needs.                                                   |
| R03 | Scope/Feature Creep            | As we are developing a game, we run the risk of encountering scope creep, i.e. continuously planning new features, which would result in us eventually not being able to keep up with the workload.                                          | We created a detailed project plan with long-term and short-term planning, along with well defined milestones and a clear MVP.                                                                                       |
| R04 | Bad Code Quality               | In large codebases it is easy to end up with badly designed systems that don't fit together with the rest of the code. This would hinder further development.                                                                                | We do mandatory code reviews before changes can be merged. Additionally, we will set up CI/CD pipelines to automatically run code quality and formatting checks on pull requests.                                    |
| R05 | Team Member is MIA             | A team member might be MIA for an extended period of time due to various reasons, such as an illness or damaged hardware. This would reduce our capacity and slow down development.                                                          | We can't reduce the possibility of this occurring, but we can distribute work across the remaining team members. We also avoid knowledge monopolies and always aim to share discoveries with the other team members. |
| R06 | Personal Conflicts             | There is always the potential for personal conflicts within the team, which would complicate development.                                                                                                                                    | We openly communicate any interpersonal issues and try to resolve conflicts as early as possible.                                                                                                                    |
| R07 | Data Loss                      | There is the potential of local storage media failing, resulting in significant data loss.                                                                                                                                                   | We mitigate this by frequently committing our changes to GitHub, and keeping untracked files in cloud based locations.                                                                                               |
| R08 | Workload Underestimation       | We might be underestimating the amount of work required for specific features and tasks.                                                                                                                                                     | We plan with a buffer and try to be conservative with our estimates. We also try to break down large tasks into smaller ones, which are easier to estimate.                                                          |

**Table 42:** Risk Analysis

### 7.4.1 Risk Matrix

The below table shows the risk assessment of the identified risks, before any mitigation strategies are applied. The risks are categorized based on their probability of occurrence and their potential severity.

| Probability | Severity   |          |          |              |
|-------------|------------|----------|----------|--------------|
|             | Negligible | Marginal | Critical | Catastrophic |
| Certain     |            |          |          |              |
| Likely      |            |          | R04      | R01          |
| Possible    |            | R02/R06  |          | R03/R08      |
| Unlikely    |            |          |          | R05          |
| Rare        |            |          |          | R07          |
| None        |            |          |          |              |

Table 43: Risk Matrix

## 7.4.2 Risk Matrix (Mitigated)

The below table shows the risk assessment of the identified risks, after mitigation strategies have been applied.

| Probability | Severity   |          |             |              |
|-------------|------------|----------|-------------|--------------|
|             | Negligible | Marginal | Critical    | Catastrophic |
| Certain     |            |          |             |              |
| Likely      |            |          |             |              |
| Possible    |            |          |             |              |
| Unlikely    |            | R06      | R01/R04/R05 | R08          |
| Rare        | R07        | R02      |             |              |
| None        |            |          |             | R03          |

Table 44: Risk Matrix (Mitigated)

# 8 Time Tracking Report

## 8.1 Time Tracking

Every member tracks time on Jira on the corresponding task/story. Time spent in meetings is documented in the meeting protocols. The time is logged in 15 minute blocks in the format e.g. 3h 15m.

## 8.2 Time Statistics

The time is grouped by member and by sprint. Per sprint and person the time is rounded to the nearest hour.

| Sprint | Cedric Cathomas | Nathanael Fässler | Yoris Kucera | Dominik Wyss | Total |
|--------|-----------------|-------------------|--------------|--------------|-------|
| 1      | 23h             | 29h               | 21h          | 24h          | 97h   |
| 2      |                 |                   |              |              |       |
| 3      |                 |                   |              |              |       |
| 4      |                 |                   |              |              |       |
| 5      |                 |                   |              |              |       |
| 6      |                 |                   |              |              |       |
| 7      |                 |                   |              |              |       |
| Total  | 23h             | 29h               | 21h          | 24h          | 97h   |

**Table 45:** Time Statistics Sprint 1

# 9 Personal Reports

# 10 Meeting Minutes

## 10.1 Weekly Scrum / 28.02.2026

| Field        | Value                                         |
|--------------|-----------------------------------------------|
| Date         | 28.02.2026                                    |
| Time         | 11:00–14:00                                   |
| Sprint       | Sprint 1                                      |
| Location     | Online                                        |
| Participants | Nathanael Fässler, Dominik Wyss, Yoris Kucera |
| Excused      | Cedric Cathomas                               |
| Absent       | -                                             |
| Meeting lead | Yoris Kucera                                  |
| Minutes by   | Nathanael Fässler                             |
| Links        | -                                             |

### 10.1.1 Agenda

| Agenda item         | Owner     | Result / Decision                                                                                                                                                                                                                                                                                                                                                 |
|---------------------|-----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| GitHub Setup        | Yoris     | <ul style="list-style-type: none"> <li>We switched from GitLab to GitHub</li> </ul> <p>The following rules were discussed:</p> <ul style="list-style-type: none"> <li>Pull Request requires one approval that is not the same person as the author.</li> <li>Merging Methods are Merge and Squash.</li> <li>Every Jira Task corresponds to one branch.</li> </ul> |
| CICD Pipeline Setup | Dominik   | <ul style="list-style-type: none"> <li>Web build and .exe build succeeded.</li> <li>Pages Architecture with main and dev branches decided</li> </ul>                                                                                                                                                                                                              |
| Timetracking        | Dominik   | <ul style="list-style-type: none"> <li>We won't use GitLab timetracking because we switched to GitHub</li> <li>Timetracking is in Jira: Everyone tracks time for individual Tasks/Stories.</li> </ul>                                                                                                                                                             |
| Unity/Coding Setup  | Yoris     | Coding Guidelines decided                                                                                                                                                                                                                                                                                                                                         |
| Steam Release       | Nathanael | <ul style="list-style-type: none"> <li>Registered on Steamworks</li> <li>Up to 30 days waiting time until application is processed</li> <li>Maybe we need to make a contract with OST.</li> </ul>                                                                                                                                                                 |
| Testing             | Nathanael | Unit-Test and Integration-Test Strategy discussed.                                                                                                                                                                                                                                                                                                                |
| Long/Shorttermplan  | Nathanael | <ul style="list-style-type: none"> <li>Hierarchy: There are Epics which contain Stories.</li> <li>Everything that is not a game-feature is a task.</li> <li>Longtermplan with epics according to the Jira Timeline</li> <li>Shorttermplan per sprint on the Jira board</li> </ul>                                                                                 |

|  |  |                                                                                                                                                                                                                                                                                       |
|--|--|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  |  | <ul style="list-style-type: none"><li>• Priority is defined by the order in the backlog in Jira</li><li>• MVP is defined</li><li>• We will estimate Tasks/Stories with Story-points</li><li>• On Jira we can create Bug-Tickets which we will not estimate with Storypoints</li></ul> |
|--|--|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

### 10.1.2 Todos

| Todo                           | Owner     | Due        |
|--------------------------------|-----------|------------|
| Finish CICD Pipeline           | Dominik   | 06.03.2026 |
| Ask Advisor about contract     | Nathanael | 04.03.2026 |
| Ask Advisor about project plan | Nathanael | 04.03.2026 |

## 10.2 Weekly Scrum and Sprint Retrospective / 06.03.2026

| Field        | Value                                                          |
|--------------|----------------------------------------------------------------|
| Date         | 06.03.2026                                                     |
| Time         | 14:00-16:00                                                    |
| Sprint       | Sprint 1                                                       |
| Location     | Online                                                         |
| Participants | Nathanael Fässler, Dominik Wyss, Yoris Kucera, Cedric Cathomas |
| Excused      | -                                                              |
| Absent       | -                                                              |
| Meeting lead | Nathanael Fässler                                              |
| Minutes by   | Cedric Cathomas                                                |
| Links        | -                                                              |

### 10.2.1 Agenda

| Agenda item                                                                                                  | Owner             | Result / Decision                                                                                                                                                                                |
|--------------------------------------------------------------------------------------------------------------|-------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Board Review                                                                                                 | All               | <ul style="list-style-type: none"> <li>Walked through all current board tasks</li> <li>Updated status on completed tasks</li> </ul>                                                              |
| Done Tasks and New Functionalities                                                                           | All               | <ul style="list-style-type: none"> <li>Reviewed completed work from Sprint 1</li> <li>Discussed new functionalities and rules</li> </ul>                                                         |
| Open Questions                                                                                               | All               | <ul style="list-style-type: none"> <li>Discussed open questions regarding project setup and workflow</li> </ul>                                                                                  |
| Unity Graphics in Repository                                                                                 | Nathanael Fässler | <ul style="list-style-type: none"> <li>Only .fbx files are to be committed to the repository for 3D graphics in Unity</li> </ul>                                                                 |
| Documentation Versioning                                                                                     | Cedric Cathomas   | <ul style="list-style-type: none"> <li>Tags for documentation versions start with "d" (e.g. d1.1)</li> <li>Version is updated automatically on pull request to main via GitHub Action</li> </ul> |
| How do we manage smaller tasks that are not worth creating a new task on jira. (small documentation changes) | Dominik Wyss      | We create ticket for small backlog tasks that do not need an own Jira item so we can reference it on our pull request and use the name of that ticket for the branch                             |
| Decide if we need a separate pdf for protocols                                                               | Cedric Cathomas   | We decided that there's no use in having a separate page just for meeting minutes, so we have one thing less to worry about                                                                      |

### 10.2.2 Sprint 1 Retrospective

#### 10.2.2.1 What Went Well

- Good and fast communication
- Good planning on Jira
- Good preplanning visible
- Product Owner did a good job
- Good and understandable selection of planning and working tools
- Pipeline and GitHub are working
- Typst is set up
- General setup went very well

### 10.2.2.2 What Didn't Go Well

- Problems with GitLab
- Didn't have enough time for some tasks

### 10.2.2.3 Improvements

- Better time management
- Better definitions of done
- Use GitHub from the beginning

## 10.2.3 Todos

| Todo                                                                              | Owner             | Due        |
|-----------------------------------------------------------------------------------|-------------------|------------|
| Create ticket for small backlog tasks that do not need an own Jira item           | Nathanael Fässler | 07.03.2026 |
| Create all table references so documentation looks nice and page breaks are clean | Cedric Cathomas   | 07.03.2026 |
| Adapt milestone dates to single dates (e.g. 16.02.2026 instead of a time frame)   | Cedric Cathomas   | 07.03.2026 |
| clean up documentation (spaces that are too big)                                  | Cedric Cathomas   | 07.03.2026 |
| Change date format in documentation to use dots                                   | Cedric Cathomas   | 07.03.2026 |
| Create task for combined build and documentation pipeline                         | Nathanael Fässler | 07.03.2026 |
| Use figures for images so they appear in the list of figures                      | Nathanael Fässler | 07.03.2026 |
| remove seperate meeting minutes pdf typst code                                    | Cedric Cathomas   | 07.03.2026 |

## 10.3 Sprint Planning / 06.03.2026

| Field        | Value                                                          |
|--------------|----------------------------------------------------------------|
| Date         | 06.03.2026                                                     |
| Time         | 16:00–18:00                                                    |
| Sprint       | Sprint 2                                                       |
| Location     | Online                                                         |
| Participants | Nathanael Fässler, Dominik Wyss, Yoris Kucera, Cedric Cathomas |
| Excused      | -                                                              |
| Absent       | -                                                              |
| Meeting lead | Yoris Kucera                                                   |
| Minutes by   | Nathanael Fässler                                              |
| Links        | -                                                              |

### 10.3.1 Sprint goal

- Build first working prototype.
- Define Functional Requirements and Non-Functional Requirements.
- Create Domain Model

## 10.4 Sprint Review (Advisor) / 10.03.2026

| Field        | Value                                                                        |
|--------------|------------------------------------------------------------------------------|
| Date         | 10.03.2026                                                                   |
| Time         | 15:00–16:00                                                                  |
| Sprint       | Sprint 2                                                                     |
| Location     | Building 8                                                                   |
| Participants | Thomas Bocek, Dominik Wyss, Yoris Kucera, Nathanael Fässler, Cedric Cathomas |
| Excused      | -                                                                            |
| Absent       | -                                                                            |
| Meeting lead | Yoris Kucera                                                                 |
| Minutes by   | Nathanael Fässler                                                            |
| Links        | -                                                                            |

### 10.4.1 Agenda

| Agenda item                               | Owner               | Result / Decision                                                                                                                                                                                                                                                       |
|-------------------------------------------|---------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Status of Work                            | Yoris               | Feedback was good                                                                                                                                                                                                                                                       |
| Project Plan                              | Yoris               | Good impression. Advisor liked that we asked him at 02.03.2026 on our initiative                                                                                                                                                                                        |
| Showcase Pipeline                         | Dominik             | Positive that Dominik sent the pipeline beforehand                                                                                                                                                                                                                      |
| Feedback                                  | Thomas              | Overall very good                                                                                                                                                                                                                                                       |
| Questions about Grading & Review Schedule | Dominik & Nathanael | <ul style="list-style-type: none"> <li>To get a better grade we have to do more than what is required on the checklist.</li> <li>The group can choose when the Review Meetings take place.</li> <li>Meeting 5 has to be before 05.06.2026 (final submission)</li> </ul> |

### 10.4.2 Todos

| Todo                                                                 | Owner           | Due        |
|----------------------------------------------------------------------|-----------------|------------|
| Update README on GitLab to indicate that project has moved to GitHub | Yoris           | 20.03.2026 |
| Configure Pipeline to build documentation                            | Not assigned    | 20.03.2026 |
| Add AGENTS.md                                                        | Cedric Cathomas | 20.03.2026 |

### 10.4.3 Next dates

| Next date               | When             |
|-------------------------|------------------|
| Review 2 - Requirements | 24.03.2026 15:00 |

## 10.5 Weekly Scrum / 13.03.2026

| Field        | Value                                                          |
|--------------|----------------------------------------------------------------|
| Date         | 13.03.2026                                                     |
| Time         | 14:30-15:30                                                    |
| Sprint       | Sprint 2                                                       |
| Location     | Online                                                         |
| Participants | Cedric Cathomas, Nathanael Fässler, Dominik Wyss, Yoris Kucera |
| Excused      | -                                                              |
| Absent       | -                                                              |
| Meeting lead | Dominik Wyss                                                   |
| Minutes by   | Yoris Kucera                                                   |
| Links        | -                                                              |

### 10.5.1 Agenda

| Agenda item                  | Owner                     | Result / Decision                                                                                                                                                                                                                                                                                                                                           |
|------------------------------|---------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Time Tracking                | Dominik Wyss              | <ul style="list-style-type: none"> <li>New time tracking report system integrated in Jira (Worklogs Report)</li> <li>Task does not need to be assigned for time to be tracked</li> </ul>                                                                                                                                                                    |
| Documentation Build Pipeline | Yoris Kucera              | <ul style="list-style-type: none"> <li>Pipeline for building documentation on GitHub Actions is set up and working</li> <li>Automatically deployed to pages on push to dev and main branches.</li> </ul>                                                                                                                                                    |
| Domain Model                 | Yoris Kucera              | <ul style="list-style-type: none"> <li>First version of domain model created</li> <li>State machine for enemy created</li> </ul>                                                                                                                                                                                                                            |
| Functional Requirements      | Nathanael Fässler         | <ul style="list-style-type: none"> <li>Documented Functional Requirements, Actors and Personas</li> <li>Documented Use Cases</li> <li>Use Cases formatting could be improved</li> </ul>                                                                                                                                                                     |
| Player and Camera            | Nathanael Fässler         | <ul style="list-style-type: none"> <li>Player controller implemented and working, controls with WASD</li> <li>Camera implemented and working, controls with mouse movement</li> <li>Camera follows player and rotates around it, fixed height and distance</li> </ul>                                                                                       |
| Non-Functional Requirements  | Cedric Cathomas           | <ul style="list-style-type: none"> <li>Documented Non-Functional Requirements</li> <li>Priority Matrix for NFRs</li> <li>NFRs grouped by category and numbered (e.g. NFR101, NFR102, NFR403, etc.)</li> </ul>                                                                                                                                               |
| GitHub CI/CD Failing         | Dominik Wyss/Yoris Kucera | <ul style="list-style-type: none"> <li>Pipeline fails because of lack of disk space on the runner</li> <li>Possible solutions:               <ul style="list-style-type: none"> <li>▸ Clean up even more storage</li> <li>▸ Make repository public</li> <li>▸ Use self-hosted runner with more storage</li> <li>▸ Rent server at OST</li> </ul> </li> </ul> |

|             |                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|-------------|----------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| UI Sketches | Dominik Wyss/<br>Cedric Cathomas | <ul style="list-style-type: none"> <li>• Sketches for Main Menu               <ul style="list-style-type: none"> <li>▸ Scene from the game shown in the background</li> <li>▸ Buttons for New Game, Continue, Save Files, Settings, Exit</li> </ul> </li> <li>• Sketch for New Game menu               <ul style="list-style-type: none"> <li>▸ Give name to world</li> </ul> </li> <li>• Sketch for Save File Menu               <ul style="list-style-type: none"> <li>▸ List of save files with name and date</li> <li>▸ Button to delete save file</li> <li>▸ Show glowcore level?</li> <li>▸ Separate list for auto saves and manual saves?</li> </ul> </li> <li>• Sketch for Settings Menu               <ul style="list-style-type: none"> <li>▸ Sliders for music and sound effects volume</li> <li>▸ Slider for mouse sensitivity?</li> <li>▸ Graphics options?</li> </ul> </li> <li>• Sketch for Pause Menu</li> </ul> |
|-------------|----------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

## 10.5.2 Todos

| Todo                                        | Owner           | Due        |
|---------------------------------------------|-----------------|------------|
| Fix WebGL Build Pipeline                    | Yoris Kucera    | 20.03.2026 |
| Unify NFRs, FRs and Domain Model            | Yoris Kucera    | 20.03.2026 |
| Adjust README on GitLab                     | Dominik Wyss    | 20.03.2026 |
| Agents.md                                   | Cedric Cathomas | 20.03.2026 |
| Explicitly address review feedback in Teams | Dominik Wyss    | 20.03.2026 |

## 10.6 Weekly Scrum and Sprint Retrospective / DD.MM.YYYY

| Field        | Value            |
|--------------|------------------|
| Date         | DD.MM.YYYY       |
| Time         | 15:00–16:00      |
| Sprint       | Sprint 3         |
| Location     | Room / Online    |
| Participants | Team A, Team B   |
| Excused      | -                |
| Absent       | -                |
| Meeting lead | Scrum Master     |
| Minutes by   | Name             |
| Links        | Retro board link |

### 10.6.1 What went well

- ...

### 10.6.2 What didn't

- ...

### 10.6.3 Todos

| Todo                                     | Owner | Due        |
|------------------------------------------|-------|------------|
| Introduce PR review rule: 1 reviewer min | Name  | DD.MM.YYYY |
| Add DoD checklist to every ticket        | Name  | DD.MM.YYYY |
| Open                                     | Name  | DD.MM.YYYY |

### 10.6.4 Retro decisions

- Process decision: ...

## 10.7 Sprint Planning / DD.MM.YYYY

| Field        | Value          |
|--------------|----------------|
| Date         | DD.MM.YYYY     |
| Time         | 16:00–17:00    |
| Sprint       | Sprint 4       |
| Location     | Room / Online  |
| Participants | Team A, Team B |
| Excused      | -              |
| Absent       | -              |
| Meeting lead | PO / SM        |
| Minutes by   | Name           |
| Links        | Backlog link   |

### 10.7.1 Sprint goal

- ...

### 10.7.2 Selected backlog (commitment)

| ID    | Item                 | Est. | Owner |
|-------|----------------------|------|-------|
| T-123 | Implement login flow | 5    | Name  |
| T-124 | Add CI pipeline      | 3    | Name  |

### 10.7.3 Risks / Dependencies

- Dependency: ...
- Risk: ...

### 10.7.4 Todos

| Todo                | Owner | Due        |
|---------------------|-------|------------|
| Prepare demo script | Name  | YYYY-MM-DD |
| Open                |       |            |



# Bibliography

# List of Figures

|                                            |           |
|--------------------------------------------|-----------|
| <b>Image 1 Use Case Diagram</b>            | <b>6</b>  |
| <b>Image 2 Domain Model Diagram</b>        | <b>18</b> |
| <b>Image 3 Enemy State Machine Diagram</b> | <b>19</b> |
| <b>Image 4 Jira Timeline</b>               | <b>31</b> |
| <b>Image 5 Inital Backlog</b>              | <b>32</b> |
| <b>Image 6 Sprint 1 Backlog</b>            | <b>33</b> |

# List of Tables

|                 |                                                                   |           |
|-----------------|-------------------------------------------------------------------|-----------|
| <b>Table 1</b>  | <b>Glossary</b>                                                   | <b>4</b>  |
| <b>Tab. 2</b>   | <b>UC01 - Install Game</b>                                        | <b>7</b>  |
| <b>Tab. 3</b>   | <b>UC02 - Save Game</b>                                           | <b>7</b>  |
| <b>Tab. 4</b>   | <b>UC03 - Break Nodes</b>                                         | <b>7</b>  |
| <b>Tab. 5</b>   | <b>UC04 - Expand Map</b>                                          | <b>7</b>  |
| <b>Tab. 6</b>   | <b>UC05 - Store Items</b>                                         | <b>7</b>  |
| <b>Tab. 7</b>   | <b>UC06 - Craft Items</b>                                         | <b>8</b>  |
| <b>Tab. 8</b>   | <b>UC07 - Build Nodes</b>                                         | <b>8</b>  |
| <b>Tab. 9</b>   | <b>UC08 - Experience Story</b>                                    | <b>8</b>  |
| <b>Tab. 10</b>  | <b>UC09 - Fight Enemies</b>                                       | <b>8</b>  |
| <b>Tab. 11</b>  | <b>UC10 - Automate Resource Gathering</b>                         | <b>8</b>  |
| <b>Tab. 12</b>  | <b>UC11 - Change Settings</b>                                     | <b>9</b>  |
| <b>Table 13</b> | <b>NFR Overview – All Non-Functional Requirements at a Glance</b> | <b>10</b> |
| <b>Table 14</b> | <b>NFR Priority Matrix – MVP Relevance vs. Importance</b>         | <b>11</b> |
| <b>Table 15</b> | <b>NFR101 – World Load Time</b>                                   | <b>11</b> |
| <b>Table 16</b> | <b>NFR102 – In-Game Transition Time</b>                           | <b>11</b> |
| <b>Table 17</b> | <b>NFR103 – Frame Rate</b>                                        | <b>12</b> |
| <b>Table 18</b> | <b>NFR104 – Save File Size</b>                                    | <b>12</b> |
| <b>Table 19</b> | <b>NFR105 – Minimal Hardware Requirements</b>                     | <b>12</b> |
| <b>Table 20</b> | <b>NFR201 – Player Restriction Visibility</b>                     | <b>13</b> |
| <b>Table 21</b> | <b>NFR202 – Light Upgrade Progress Indication</b>                 | <b>13</b> |
| <b>Table 22</b> | <b>NFR203 – New Player Learnability</b>                           | <b>13</b> |
| <b>Table 23</b> | <b>NFR204 – Input Support</b>                                     | <b>13</b> |
| <b>Table 24</b> | <b>NFR205 – Game Language</b>                                     | <b>14</b> |
| <b>Table 25</b> | <b>NFR206 – Visual Style Consistency</b>                          | <b>14</b> |
| <b>Table 26</b> | <b>NFR301 – Save File Portability</b>                             | <b>14</b> |

|                                                        |           |
|--------------------------------------------------------|-----------|
| <b>Table 27 NFR302 – Game Progress Persistence</b>     | <b>15</b> |
| <b>Table 28 NFR303 – Save Data Integrity</b>           | <b>15</b> |
| <b>Table 29 NFR401 – Extensibility of Game Systems</b> | <b>15</b> |
| <b>Table 30 NFR402 – Automated Test Coverage</b>       | <b>16</b> |
| <b>Table 31 NFR501 – Multi-Platform Support</b>        | <b>16</b> |
| <b>Table 32 NFR502 – Engine and Rendering Pipeline</b> | <b>16</b> |
| <b>Table 33 NFR503 – Display Resolution Support</b>    | <b>17</b> |
| <b>Table 34 Team availability for review meetings</b>  | <b>23</b> |
| <b>Table 35 Project Management Roles</b>               | <b>25</b> |
| <b>Table 36 Developer Roles</b>                        | <b>26</b> |
| <b>Table 37 Planned Meetings Overview</b>              | <b>27</b> |
| <b>Table 38 Planning Tools</b>                         | <b>27</b> |
| <b>Table 39 Work Items</b>                             | <b>28</b> |
| <b>Table 40 Milestones</b>                             | <b>29</b> |
| <b>Table 41 Epics</b>                                  | <b>30</b> |
| <b>Table 42 Risk Analysis</b>                          | <b>34</b> |
| <b>Table 43 Risk Matrix</b>                            | <b>35</b> |
| <b>Table 44 Risk Matrix (Mitigated)</b>                | <b>35</b> |
| <b>Table 45 Time Statistics Sprint 1</b>               | <b>36</b> |

# Code Listings